Unlock the code: acceleration

07 C

DIRECTIONS · Solve each station. Copy the letter beside your answer into the lock.

STATION 1

A cart goes from 0 to 8 m/s in 4 s. Find a.

(M) 4 m/s² (C) 2 m/s² (R) 32 m/s²

STATION 2

Velocity drops from 12 to 4 m/s in 2 s. Find a.

(P) +4 m/s² (T) −8 m/s² (H) −4 m/s²

STATION 3

v = +5 m/s and a = −2 m/s². The object is...

(E) speeding up (A) slowing down

STATION 4

v = −6 m/s and a = −3 m/s². The object is...

(L) slowing down (N) speeding up

STATION 5

v_0 = 2 m/s, a = 3 m/s², t = 4 s. Use $v = v_0 + at$.

(D) 12 m/s (G) 14 m/s (S) 9 m/s

STATION 6

A car goes 0 → 27 m/s in 6 s. Find a. ($g \approx 9.8$ m/s²)

(O) 162 m/s² (E) 4.5 m/s² (U) 21 m/s²

LOCK [1] [2] [3] [4] [5] [6]

Hint: what acceleration measures in velocity.

R4 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

UNIT 1 · DAY 7 · R4 · SEI · PARTNER TALK

Talk it through: acceleration

Language goal: I can **explain** speeding up and slowing down.

DIRECTIONS · Partner A and B read their lines aloud. Listen, then write. Switch each round.

ROUND 1 · A CAR: 0 → 12 m/s IN 4 s



0 → 12 m/s in 4 s

A ASKS "What is the acceleration of the car?"

B SAYS "The change in velocity is ____. I divide by ____. So a = ____."

A CHECKS "So you're saying the car gains ____ m/s every second. I agree / disagree because ____."

ROUND 2 · SWITCH · A BIKE BRAKES



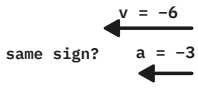
bike brakes

B ASKS "Is the bike speeding up or slowing down? How do you know?"

A SAYS "The bike is ____ because the gaps ____. Its acceleration points ____ the motion."

B ADDS "I want to add that braking is still acceleration because ____."

ROUND 3 · BOTH · v = −6 m/s, a = −3 m/s²



A "I think it is ____ because the signs are ____."

B "I agree / I see it differently. The minus sign means ____."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

Acceleration means…

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

R4 (1) (2) (3)

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Settle the argument

07 I

DIRECTIONS · Shade the bubble of the person who is right. Give evidence, then fix the wrong claim.

ROUND 1

ANA
"Negative acceleration means slowing down."

BEN
"Not always — it depends on the sign of the velocity too."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 2

KAI
"A car going a steady 20 m/s on a curve isn't accelerating."

LU
"It is — its direction, so its velocity, is changing."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 3

MAYA
"2 → 10 m/s in 4 s gives a = 10 ÷ 4 = 2.5 m/s²."

RAJ
"You need the change: a = (10 − 2) ÷ 4."

Who's right? _____ · Evidence: _____

Fixed claim: _____

YOUR TURN · WRITE A TRAP

Write a claim about today's idea that sounds right but is wrong. Trade and settle it.

R4 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Say it, then write it

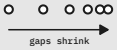
Language goal: I can **explain** acceleration with **because**.



accelerate
acelerar · acelerar
to change velocity



speed up
acelerar · acelerar
go faster · gaps grow



slow down
frenar · desacelerar
go slower · gaps shrink



change in
cambio de · variação de
Δ = end − start

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: faster · slower · velocity · second · change

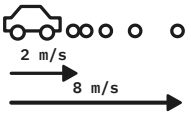
Acceleration tells how much the ____ changes each ____. When a car goes ____, it speeds up. When it goes ____, it slows down. Both are a ____ in velocity.

2 · SAY IT TO A PARTNER, THEN WRITE IT



The car is ____ because the gaps ____ .
v and a have ____ signs, so the car is ____ .

3 · SOLVE WITH THE STEPS



A bike goes from 2 m/s to 8 m/s in 3 seconds.

STEP 1 · CHANGE
____ − ____ = ____ m/s

STEP 2 · DIVIDE
____ ÷ ____ = ____ m/s²

STEP 3 · SAY IT
"The bike gains ____ m/s every second."

4 · EXPLAIN IT · CHOOSE A FRAME

☐ Braking is acceleration because ____ .
☐ I know the car speeds up because ____ .
☐ First the velocity is ____, then ____, so a is ____ .

I said it ☐ · I wrote it ☐

R4 1 2 3

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☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

100 m final – the first 5 seconds

● LIVE

You're the broadcast analyst. Made-up practice race.

07 X

SPLIT DATA

t (s)	v (m/s)	a (m/s²)
0	0	–
1	4.0	
2	7.5	
3	10.0	
4	11.0	
5	11.0	

SLOW-MO v–t GRAPH

STAT 1 · BIGGEST a

Which second? ____

STAT 2 · a REACHES 0

When? ____ Still moving? ____

STAT 3 · AVERAGE a, 0–4 s

a = ____ · vs. g ≈ 9.8: ____

COMMENTARY BOOTH

"Out of the blocks she accelerates at ____ – a gain of ____ every second! By second 4 she holds a top speed of ____, so her acceleration is ____ … but she is still ____!"

R4 1 2 3

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☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

UNIT 1 · DAY 7 · R4 · LOOP CHAIN

I have... who has? Acceleration

07 G

DIRECTIONS · Start at ★. Solve, find the tile that "has" your answer, draw an arrow, repeat.

★ START — **A**

WHO HAS…

0 → 8 m/s in 4 s. What is a?

I have **14 m/s** — **B**

WHO HAS…

0 → 27 m/s in 6 s. What is a?

I have **−4 m/s²** — **C**

WHO HAS…

v = +5, a = −2. Speeding up or slowing?

I have **m/s²** — **D**

WHO HAS…

v = −6, a = −3. Speeding up or slowing?

I have **2 m/s²** — **E**

WHO HAS…

12 → 4 m/s in 2 s. What is a?

I have **speeding up** — **F**

WHO HAS…

Steady 20 m/s around a curve. Accelerating?

I have **slowing down** — **G**

WHO HAS…

v0 = 2, a = 3, t = 4 s. What is v?

I have **4.5 m/s²** — **H**

WHO HAS…

The units of acceleration?

I have **yes — direction changes** — **I**

END

MY PATH: A → ___ → ___ → ___ → ___ → ___ → ___ → ___ → END

R4 1 2 3

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Comic: acceleration is any change

07 CS

YOUR READER
A 4th grader who thinks acceleration only means speeding up.

Stick figures are fine. The physics gets checked.

MUST-HAVES · check it off, then color its circle

- ☐ ◌ **PURPLE** — A velocity arrow in every panel, with lengths that change
- ☐ ◌ **GREEN** — The speeding-up panel's border
- ☐ ◌ **YELLOW** — The steady panel's border
- ☐ ◌ **RED** — The slowing-down panel's border

1 · SPEEDING UP *A bike rider pedals harder.*

CAPTION _____

2 · STEADY *The rider cruises at the same velocity.*

CAPTION _____

3 · SLOWING DOWN *The rider brakes.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **acceleration · velocity · speeding up · slowing down**

R4 1 2 3

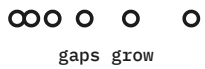
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Doodle Notes · acceleration

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Acceleration · aceleración (ES) · aceleração (PT)



gaps grow

TODAY'S CASE

At 0, 1 and 2 seconds, a bike is moving right at 6, 4 and 2 m/s.

01 · SKETCH AND LABEL

Draw three purple velocity arrows to scale. Label each speed and time.

02 · CHANGE THE PICTURE

Draw the change in velocity between two arrows. Show how the bike can still move right as it slows down.

Annotate your sketches. Circle the mark that best shows acceleration. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

The velocity points ___ while the change points ...

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☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Frayer Model · acceleration

I can explain a term by comparing an example with a nonexample.

Name: _____ My home language: _____

Acceleration · aceleración (ES) · aceleração (PT)

A · A bike moving straight at an unchanging speed has acceleration.

B · A bike turning at a steady speed changes its velocity.

01 · DEFINITION

What does acceleration mean in your own words? Use the notes or start in your home language.

02 · CHARACTERISTICS

What must be true? Label a small sketch to show one feature.

03 · EXAMPLE

Choose A or B as your example. Sketch it and explain which feature makes it fit.

04 · NONEXAMPLE

Explain why the other card does not fit. Change one detail so that it would.

SAY IT TO A PARTNER · THEN WRITE

I chose card ___ as my example because …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
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Picture Dictionary · acceleration

I can connect each word to a picture and use it to explain this case.

Name: _____ My home language: _____



accelerate

acelerar · acelerar

to change velocity

My word or label: _____

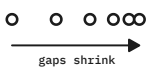


speed up

acelerar · acelerar

go faster · gaps grow

My word or label: _____

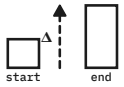


slow down

frenar · desacelerar

go slower · gaps shrink

My word or label: _____

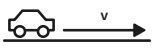


change in

cambio de · variação de

Δ = end − start

My word or label: _____

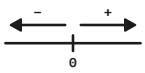


velocity

velocidad · velocidade

how fast and in which direction

My word or label: _____



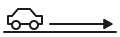
direction

dirección · direção

which way an object is moving

My word or label: _____

MATCH · WRITE THE ENGLISH WORD UNDER EACH PICTURE



_____



_____



_____

Use two words to explain the case: At 0, 1 and 2 seconds, a bike is moving right at 6, 4 and 2 m/s.

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